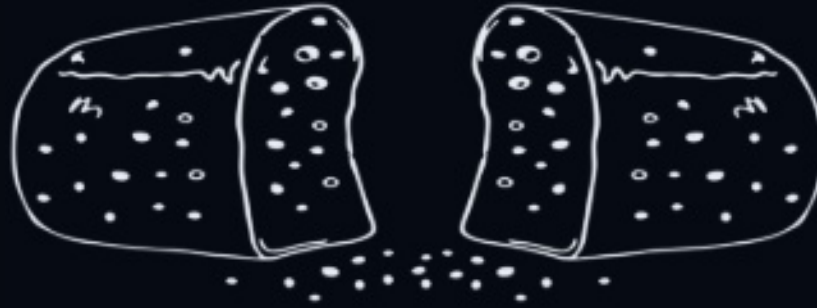




Our food supply is a black box.

Follow the crumbs back to the source –
and forward to every plate at risk.

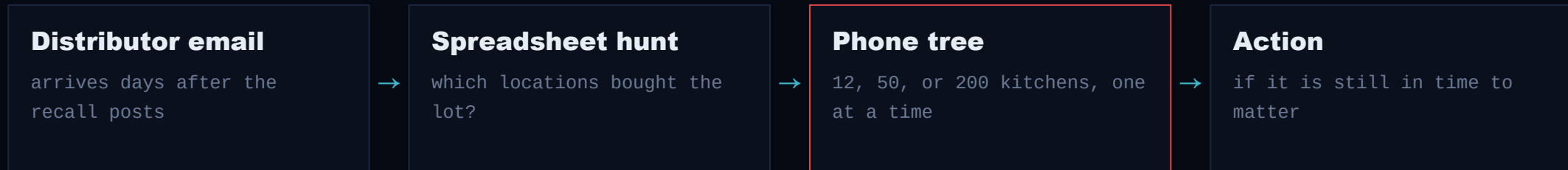
ALWAYS-ON FOOD TRACEABILITY

· ON-PREM

· NVIDIA GB10

How it works today

distributor email → days later → spreadsheet hunt → phone tree



THE STAKES ARE REAL MONEY

\$0.04M–\$1.1M

median per-recall cost, per restaurant
firm

~\$10M

average direct recall cost for grocers
(FMI/GMA)

52%

of major recalls exceed \$10M in total
impact

The reputational tail is far larger.

BREADCRUMBS

4 seconds

on a box in the back office,
supplier list never leaving the building

TODAY

2 days

email, spreadsheets, and a phone tree
across every location

That's the difference between a recall and an outbreak.

Why it has to run local

A food business's supplier list, purchase volumes and margins are trade secrets.

No one POSTs that to a cloud LLM to get a recall alert.

The alert isn't worth the leak.

Local isn't a constraint we're excusing – it's the only shape the customer will accept.

FSMA 204

FDA Food Traceability Rule

MANDATORY
JULY 20, 2028

Lot-level traceability becomes a legal recordkeeping mandate – so this data is about to exist in every back office whether they like it or not.

Live demo

a real NYC food business · 12 sites · their actual buildings

12

sites in the portfolio

0

bytes leaving the box

29,310

FDA recalls indexed locally

17,204

geocoded sites with violation counts

SAY IT, DON'T WAIT TO BE ASKED

“The buildings and inspection data are real. This 12-site portfolio is an illustrative operator we modeled — I'll show you what's real underneath in a second.”

BREADCRUMBS is watching the FDA recall feed for them right now, and every byte is running on this box. No internet.

A Class I recall drops

F-0757-2022

DOLE FRESH VEGETABLES INC — iceberg lettuce

Listeria monocytogenes · Class I · firm state CA

WHAT CLASS I MEANS

“A reasonable probability of serious adverse health consequences or death.”

21 CFR 7.3 — the FDA's most serious classification.

PROVENANCE

REAL

openFDA food enforcement.

Real recall, real number — one of 29,310 in the local corpus.

Trace it forward

5 of 12

sites received the lot.
The other seven are fine – and
knowing which is the whole game.

SITE	SUPPLIER LINK → THIS LOT	VERIFY AGAINST
Site 05 — LES	Dole → iceberg	Receiving log
Site 11 — Harlem	Dole → iceberg	Receiving log
Site 03 — Midtown	Dole → iceberg	Receiving log
Site 07 — Williamsburg	Dole → iceberg	Receiving log
Site 09 — LIC	Dole → iceberg	Receiving log

MODELED — SAY IT IN THE SAME BREATH AS THE NUMBER

The supplier→site link is the one modeled layer — and that's the point. FSMA 204 is the rule that's supposed to make this traceable and doesn't yet. We model the gap the law is trying to close.

Two independent, real signals

1 — DISTRIBUTOR RECALL HISTORY

FIRM	CLASS-I	READ
Dole	205	Extensive history
Fresh Express	110	High
Sysco	24	Moderate
Jetro	0 of 1	Verified low record

REAL — openFDA firm-level recall counts

2 — SITE HANDLING RISK

SITE	CRIT	VS 4	READ
Site 05 — LES	14	+10	Elevated
Site 11 — Harlem	9	+5	Above median
Site 03 — Midtown	6	+2	Slightly above
Site 07 — Williamsburg	4	0	At median

REAL — DOHMH crit_violations (0-56, median 4)

0 of 1 is a VERIFIED LOW RECORD — never “clean”.

An absence of matched records is not proof of a safer product; it can also mean lower recall visibility. Never say “C health grade” — grades composite vermin, facility, paperwork and temperature. We use crit_violations.

What the manager actually gets

a document a health inspector, an insurer, or a plaintiff's attorney already recognises

Recall Exposure & Action Report

- A FDA recall facts – mapped 1:1 to openFDA
- B Exposure – which sites received the lot
- C Risk – two independent real signals
- D Recommended action – deterministic
- E Real vs. modeled provenance

21 CFR 7.3 · classification

21 CFR 7.49 · recall communication

TELEGRAM / SMS OUTBOX

FDA CLASS I RECALL – action needed today.

Dole iceberg lettuce (Listeria), recall F-0757-2022.

5 of your 12 sites received matching lots: Midtown,
LES, Williamsburg, LIC, Harlem.

DO: hold & segregate now; verify lot codes; do not serve.

Full report + swap: BC-2026-0822-0417 (PDF).

Drafted, not dispatched.

A human still hits go.

A dated, on-prem-generated report showing what they knew, when, and what action was directed is contemporaneous evidence of a good-faith response.

Where we work — and where we don't

volunteer this before a judge digs for it

REAL US OUTBREAK	RELEVANT?	WHY
Jif peanut butter, 2022	YES	FDA Class-I lot-coded recall — exactly our trigger. 24 real Smucker records verified.
McDonald's / Taylor Farms onions, 2024	YES	FDA-regulated produce, lot-coded — we map it to the sites that received the lot.
Yuma romaine, 2018	PARTIAL	Commodity produce → a public advisory, not a lot-coded recall. No lot to trace.
Boar's Head deli meat, 2024	NO	USDA FSIS jurisdiction — invisible to the openFDA food feed. Our labeled boundary.
Chipotle, 2015	NO	No recall was ever issued and the ingredient was never identified — nothing to trigger on.

~100% of FDA lot-coded enforcement recalls. A minority of the full universe of epidemiological outbreaks.

We sit downstream of CDC/FDA traceback and act the instant a recall posts. • Real cadence: 24 NY-distributed Class-I recalls in 2025-26 — roughly 1-2 a month.

An instrument, not an oracle

“Unset the API key — what changes?”

The narration sentence goes generic. Nothing else. Every red site, every risk score, the swap, the report — all deterministic Mongo aggregation. The LLM writes prose; it doesn't make the decision.

REAL

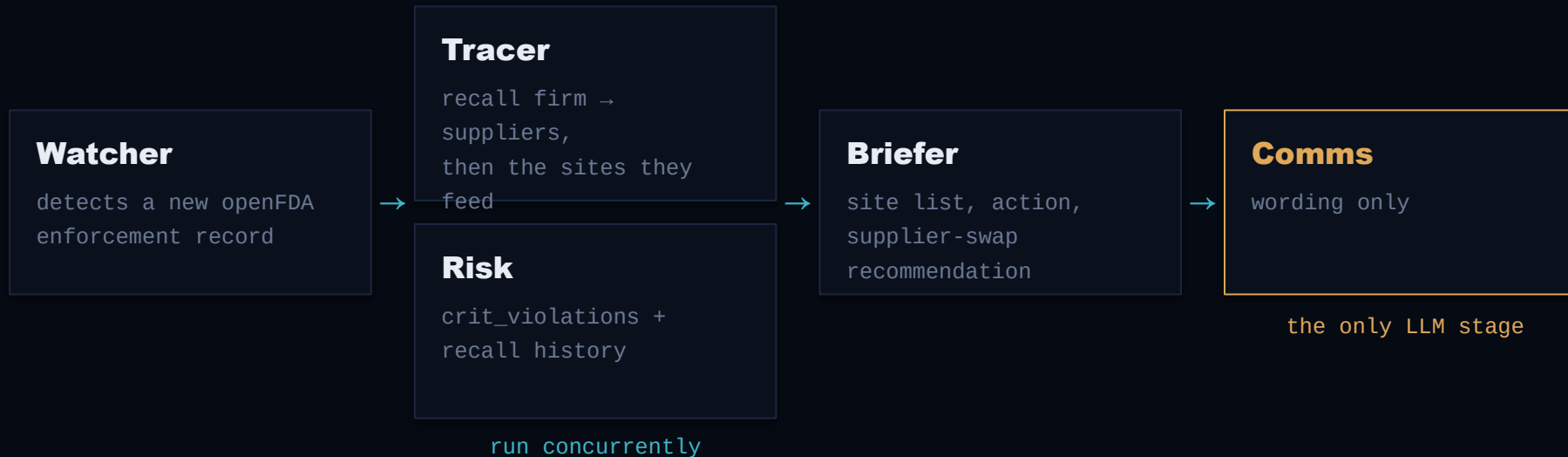
29,310 FDA recalls
17,204 geocoded sites with critical-violation counts
Distributor recall histories
LiDAR building geometry

MODELED — and labeled

Which supplier feeds which site
The illustrative 12-site portfolio

When a real food business plugs in their real supplier list, the modeled layer disappears.

Five stages. The LLM touches one.



Four of five stages are pure deterministic computation over real data — an orchestration of auditable steps, not a chain of LLM guesses.

MongoDB earns its place: 2dsphere \$near for geospatial, \$lookup/\$group to roll up exposure at query time, and a change stream that watches. A spreadsheet is a snapshot; this is a watcher.

Model: Llama-3.1-Nemotron-Nano-4B via vLLM on localhost:8000

The hinge is a regulation

and it writes our data for us

JULY 20, 2028

FSMA 204 becomes mandatory. To comply, food businesses must generate lot codes and key data elements at every critical tracking event.

Those records are exactly the edges we model today.

Feed them into the same engine and the modeled links become real provenance. No new algorithm – the same aggregation on real edges.

Wedge: recall response. Moat: the traceability system of record the law requires.

WHO BUYS IT

Regional restaurant chains

Ghost-kitchen groups

Distributors

Grocers

~\$30B by 2030 · food-traceability technology market (Grand View, 8.8% CAGR; other houses put 2030 at \$10B–\$44B)

THE NEXT REAL MILESTONE

One restaurant group. Their real purchase records. Twelve months.

That one step converts the modeled edges into real provenance and proves the platform hinge on live data.

No hockey stick – one operator, their records, on their own box. Same engine the whole way up.

The six we never overclaim

1 Recall responder, not outbreak detector

~100% of our feed (FDA lot-coded recalls); only a minority of real epidemiological outbreaks.

2 FSIS is a labeled blind spot

FDA-only. Boar's Head 2024 wouldn't appear. Pre-empt it; don't get caught.

3 Local inference is currently thin

The LLM writes one narration sentence; deterministic Mongo makes every decision.

4 Change stream isn't wired to run() yet

The Watcher observes inserts; it does not auto-trigger the pipeline. We trigger runs explicitly.

5 Supplier→site links are modeled

The FSMA-204 gap we exist to close – labeled modeled, not passed off as ground truth.

6 Telegram is draft-not-send

The Comms output is prepared, not auto-dispatched to a live operator channel.